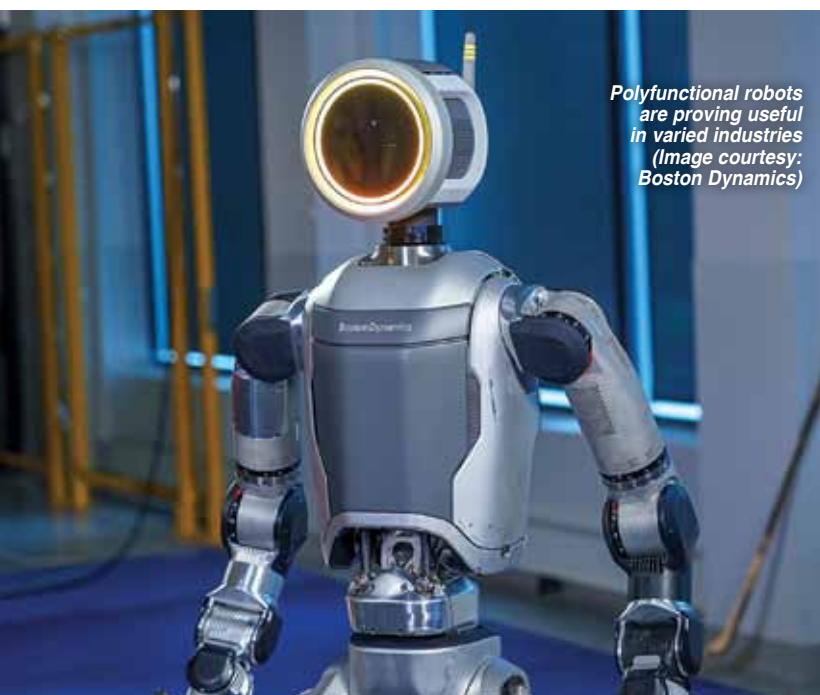


POLYFUNCTIONAL ROBOTS: One Robot For All Your Needs

Polyfunctional robots have been pinned as one of the major technological trends for 2025 by almost every market research firm. Yet, not many know what a polyfunctional robot is—and what it can do. So, we thought we would just get down to the brass tacks...



- In a sense, polyfunctional robots can be seen as the equivalent of AI agents in the robotics world

What are the constituent technologies?

- Polyfunctional robots are advanced robotic systems with multiple degrees of freedom
- They are usually collaborative robots (cobots)—that is, they can work alongside humans and other robots
- Made of modular hardware, they can dynamically switch functionalities
- They are also fitted with multiple tools and equipped with efficient mechanisms to swap tools and components to transition between tasks quickly
- They have advanced sensors and cameras that enable interaction with the environment in multiple ways, including vision, voice, and touch
- They connect to the cloud to optimise their performance, for remote control and monitoring, and software updates
- Most importantly, it is AI and machine learning (ML) that make polyfunctional robots what they are! AI and ML enable these robots to interact smoothly with the environment and with enterprise systems, autonomously make decisions, switch tasks, learn, improve, adapt, and perform predictive maintenance

What are polyfunctional robots?

- Most robots employed in today's industries are 'made for a job'—but polyfunctional robots can perform multiple tasks and are adaptable across working environments and different industries
- For example, a polyfunctional robot in defence can help with surveillance, search and rescue, bomb disposal, and reconnaissance; in a first responder team, it can assist with search and rescue, first aid, crowd management, debris cleaning, and more
- Apart from advanced robotic capabilities, they have built-in intelligence to perform multiple tasks, make decisions, and learn on-the-go



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How can businesses benefit

- Higher returns on investment (RoI) because the robot can be used in multiple ways
- For example, in a warehouse, the same robot can be deployed for item picking, packing, and/or delivery as needed. And when it is no longer required, it can be